Machine Learning Exercise 3

Exercise 1

If we set $C = 0$ in the relaxed objective of slide 3.14, what would be the optimal solution for $\boldsymbol{w}$ and b, and what would be the resulting classification rule?

Solution:
Setting $C = 0$ gives us the optimization problem: minimize

$$\frac{1}{2} \parallel \boldsymbol{w} \parallel^2$$

subject to

$$y_n(\boldsymbol{w} \cdot \boldsymbol{x}_n + b) \geq 1 - \zeta_n \quad (n = 1, \ldots, N).$$

The objective function $\parallel \boldsymbol{w} \parallel^2 /2$ would be minimized by setting $\boldsymbol{w} = \mathbf{0}$. We only have to check whether with a suitable setting of the remaining variables b, ζ_n this leads to a feasible solution. The constraints then become

$$y_n \cdot b \geq 1 - \zeta_n$$

Remembering that $y_n \in \{-1, 1\}$: setting e.g. $b = 1$ and $\zeta_n = 2$ satisfies the constraints (there are other options as well). With these $\boldsymbol{w}, b$, the linear function $\boldsymbol{w} \cdot \boldsymbol{x} + b$ evaluates to 1 for every point $\boldsymbol{x}$, i.e., all points will be classified as belonging to the positive class. There are other possible solutions for b, ζ_n that will lead to every point being classified as negative. In all cases, however, the prediction will always be the same for all points.

Exercise 2

Let t_1 be the "query" text

$$t_1 = \textit{computer science education aalborg university}$$

Let t_2 be the text content of `https://www.en.aau.dk/education/master/computer-science-it`. Assume that the vocabulary (or dictionary) with regard to which we contains just the terms

computer, education, science, university.

- What is the cosine similarity *cos-sim*(t_1, t_2)?

If we use the slightly enlarged vocabulary

computer, education, knowledge, master, science, software, technology, university.

instead, what is *cos-sim*(t_1, t_2) then?

Solution:
We first have to compute the term frequency vectors for t_1, t_2 for the two vocabularies. Searching and counting the number of occurrences of terms on the web page, we get

term	tf(t_1)	tf(t_2)
computer	1	12
education	1	3
science	1	13
university	1	10
knowledge	0	1
master	0	15
software	0	2
technology	0	2

(here also counting occurrences of "master's" as hits for "master"). If your counts differ slightly, that's no matter of concern ...

For both vocabularies we have

$$\mathrm{tf}(t_1) \cdot \mathrm{tf}(t_2) = 1 \cdot 12 + 1 \cdot 3 + 1 \cdot 13 + 1 \cdot 10 = 38.$$

With respect to the small vocabulary we get

$$\parallel \mathrm{tf}(t_1) \parallel = \sqrt{4} = 2, \qquad \parallel \mathrm{tf}(t_2) \parallel = \sqrt{12^2 + 3^2 + 13^2 + 10^2} = 20.542,$$

and so

$$\textit{cos-sim}(t_1, t_2) = \frac{38}{2 \cdot 20.542} = 0.925$$

With the bigger vocabulary, the only change is

$$\parallel \mathrm{tf}(t_2) \parallel = 25.61,$$

which leads to

$$\textit{cos-sim}(t_1, t_2) = \frac{38}{2 \cdot 25.61} = 0.741.$$

Exercise 3

Compute the kernel matrix for the 2-spectum kernel for the set of words $\{aaa, aba, baba\}$

Solution:
The relevant, non-zero, parts of the feature vectors for the three words are

	aa	ab	ba	bb
ϕ(aaa)	2	0	0	0
ϕ(aba)	0	1	1	0
ϕ(baba)	0	1	2	0

Note that it does not matter whether the underlying alphabet Σ is assumed to only contain the letters a,b, or all a-z). This gives us the kernel matrix

	aaa	aba	baba
aaa	4	0	0
aba	0	2	3
baba	0	3	5